

Operationalizing Radiant Patterns: A Refined Definition and Pattern Structure to Mitigate Deceptive Game Design Practices

Luiz Santos Filho(✉) and Ticianne Darin

Universidade Federal do Ceará, Fortaleza, Ceará, Brazil
luis.gsantosf@alu.ufc.br, ticianne@virtual.ufc.br

Abstract. Deceptive design patterns in digital games often prioritize developer interests at the expense of player well-being by manipulating behavior in potentially harmful ways. While the concept of Radiant Game Design Patterns (RPs) was introduced as a positive alternative to counteract deceptive patterns in games, its initial theoretical framing presented challenges for consistent practical application and operationalization. This paper addresses this gap by proposing two core contributions to advance ethical game design. First, we provide a refined definition of Radiant Patterns, grounded in Self-Determination Theory (SDT), which emphasizes ensuring that foundational player well-being is not significantly compromised by avoiding a decrease in players' autonomy, competence, and relatedness. Second, we present a detailed documentation structure—a practical template—for defining, instantiating, and sharing RPs that adhere to this refined definition. The application of this definition and structure is illustrated through an in-depth example of an RP. By providing these tools for operationalization, this work aims to equip the Human-Computer Interaction (HCI) community and game developers with a systematic way to collaboratively identify, develop, empirically validate, and implement ethical alternatives to deceptive patterns.

Keywords: Game Design · Self-Determination Theory · Digital Well-Being · Deceptive Patterns · Dark Patterns · Game Design Patterns · Positive Design · Healthy Game Design · Ethical Game Design and Countermeasure

1 Introduction

Deceptive design strategies have become a widespread problem in today's digital environment, significantly impacting user interactions across various platforms, including social media, online shopping [27], mobile applications [23], cookie consent interfaces [47], and video games [16]. As research on player well-being in HCI continues to advance, studies have investigated the consequences of negative design patterns in players' daily lives, especially as the digital gaming industry has evolved over the years [18,56].

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
T. Darin et al. (Eds.): WIPlay 2025, CCIS 2623, pp. 112–131, 2026.
https://doi.org/10.1007/978-3-032-01426-9_8

Although games are designed to engage players and provide an enjoyable experience [11], the gaming industry has been increasingly affected by practices that do not prioritize players' best interests [26]. Patterns of deception in digital games have been highlighted in studies over the last few years, mainly due to their negative consequences and potential to trigger disorders and malicious behavior [46,49,58].

While the HCI community has taken great steps forward in improving our understanding of dark patterns [1,31], and despite the existence of relevant studies that provide essential frameworks [12], evaluation instruments [32,34], and targeted design suggestions [42], such research predominantly emphasizes identifying issues or offering broad guidance instead of showcasing the development and thorough empirical assessment of practical and beneficial game design pattern applications.

The development of *practical* and *positive* design solutions for games requires further targeted efforts. Some steps have been taken in proposing design patterns to counteract dark patterns in general-purpose applications, such as the Bright Patterns [15]; and Fair Patterns [35]. In games, Lewis [22] presented a comprehensive set of 'Motivational Design Patterns,' grounded in psychological theory and offering a "language" for motivational design, substantiated through analytical case studies. Also, Miranda [30] developed on a new pattern concept called Radiant Patterns, offering game designers an alternative to lead the players to a state of well-being, satisfying one or more basic psychological needs. These contributions by [22] [30] provide relevant conceptual bases for actionable ethical design in games.

However, proposals such as the 'Radiant Game Design Patterns' and 'Motivational Design Patterns', although they are design patterns intended for implementation, are presented mainly at a conceptual or theoretical level. While offering principles, they often lack detailed, empirically validated examples of implementation within specific game mechanics and interfaces, as well as assessments of their performance regarding both player well-being and developer goals.

The consequence of positive patterns being largely abstract conceptual frameworks is an ongoing gap between ethical ideals and specific design strategies. Designers may grasp the concept of a 'radiant' pattern that enhances autonomy, but be unable to translate this concept when redesigning problematic mechanisms [41], such as daily quest mechanisms or in-app purchasing flows.

While theoretical frameworks are essential for advancing knowledge in the field, a critical gap exists between these concepts and their empirically validated, real-world instantiation in game design. This issue demands attention, as practitioners often lack specific, tested guidance on translating such ethical principles into concrete, effective design solutions. Thus, with this paper, we aim to help bridge this divide. We shed light on the concept of Game Radiant Patterns by refining its original definition, proposing a concrete framework for defining new patterns, implementing this framework, and guiding its practical application.

Through our work, we aim to contribute to both the game industry and the HCI community by revisiting and reshaping game design patterns that promote

players' well-being. By developing a more refined definition, we proposed a more applicable and operational concept for use by the game industry. For a better understanding of how this pattern is structured, we present a practical template. This template is exemplified by an example that enables game developers to identify and empirically validate patterns that can serve as alternatives to deceptive ones. Through this template, we invite researchers and game designers to foster the development of game design patterns grounded in ethical principles and oriented toward promoting player well-being.

2 How Deceptive Patterns in Digital Games Affects Players and Player-Experience

The negative effects of using deceptive patterns have been extensively studied in the literature [5,17,28]. Regardless of the context in which they are applied, it is clear that these patterns not only affect the player's interactions with electronic devices [17,24,57], but also negatively impact their lives out of screen [5,45]. This is no different from the context of digital games as well, where deceptive patterns are associated with issues such as the engagement-addiction dilemma [58], gaming disorder [49] and screen addiction disorder [46] - the latter two are already recognized as clinical disorders by the World Health Organization.

In his work, Zagal [59] addressed the use of deceptive patterns in digital games—referred to at the time as *dark patterns*. He not only identified groups of patterns that negatively affect players but also compiled a catalog outlining harmful design practices—essentially, a guide on "what not to use" in game design to avoid causing harm to players. Among these patterns are mechanics frequently found in famous digital games like *Genshin Impact*[1] and *Clash of Clans*[2], such as *Daily Rewards*—which encourage players to return every day and punish them for missing a day—and *Pay to Skip*—which allows players to spend money to bypass waiting periods enforced by timers. However, this raises a critical question: if a catalog of deceptive design patterns already exists, why do such patterns persist in being used?

One of the main reasons for the continued use of deceptive patterns is their ease of implementation. These patterns can be implemented at low cost [28], on a large scale [5], and still remain highly effective in capturing and manipulating players [7]. By exploiting cognitive biases [29], such patterns undermine players' autonomy, inducing individuals to spend time and money in ways that may ultimately be detrimental to their well-being [55]. Furthermore, popular titles known to incorporate deceptive patterns, such as *Genshin Impact* [19], *Final Fantasy XIV* [54], and *League of Legends* [13], serve as influential models within the industry. Following the commercial success of these games, other developers frequently attempt to replicate their mechanics and design strategies

[1] MiHoYo. 2020. Genshin Impact. Game [Android, iOS, PlayStation 4, Windows, PlayStation 5, Xbox Series X/S].

[2] Supercell. 2012. Clash of Clans [iOS, Android].

without fully understanding their negative consequences, just for financial gain and engagement.

From the player's perspective, their individual experience and level of digital literacy significantly influence how they perceive these patterns. Players who are aware of the ethics in digital games can recognize these patterns and avoid participating in games that involve them [41]. This level of literacy has a direct impact on the effectiveness of deceptive design. Many players who are unaware of the consequences of these patterns may remain active in games that use them due to the manipulation they experience, resulting in frustration and compromising their autonomy [50].

In addition to players' literacy and experience, the effectiveness of these patterns depends on players' perception of how frequently these patterns appear in games and how they are presented in the UI [25]. Due to the complexity of the consequences of deceptive patterns' use, it is necessary to study and adopt practices that are ethical and promote healthy player interactions [22].

A significant number of articles in the literature already seek to identify deceptive practices in digital games [19,48,50] and although they identify these patterns, there is still a gap in the literature regarding ways to mitigate the negative effects of deceptive game patterns and focus on a more ethically driven set of patterns.

Therefore, it is necessary to build upon the work of [59], which specifies which standards should *not be used*, and provide designers with a deeper understanding of *what should be used*. In this way, it enables game developers to adopt more ethical practices in their work. The Self-determination theory has been explored alongside human-computer interaction (HCI) as an alternative in the development of patterns that support players' well-being [4,10,30].

3 Well-Being Applied to Game Design

Over the years, video games have often been associated with negative aspects of well-being, such as aggression, reduced academic performance, and psychological disorders [36]. However, more recent studies have begun to emphasize their positive effects, particularly by referencing theories related to intrinsic motivation, such as Self-Determination Theory (SDT) [40].

According to SDT, a eudaimonic life is more likely to fulfill basic psychological needs -autonomy, competence and relatedness-, thereby enhancing well-being [38]. Psychologists have drawn on the distinction between hedonia and eudaimonia to differentiate two overlapping but distinct perspectives on well-being. Hedonic approaches define well-being primarily in terms of pleasure and happiness. In contrast, eudaimonic approaches conceptualize well-being as a multidimensional construct that goes beyond pleasure, emphasizing meaning, self-realization, and the pursuit of intrinsically valuable goals [51].

Studies grounded in SDT have shown that video games can satisfy these needs, thereby increasing intrinsic motivation. Specifically, fulfilling these needs during gameplay enhances enjoyment, the motivation to continue playing, life

satisfaction, and overall psychological well-being. When these needs are met, both the gaming experience and social interactions within games tend to be evaluated more positively [52]. On the other hand, when these needs are unmet, thwarted, or when gameplay is driven by external pressures, it is linked to diminished psychological functioning [37]. In other words, the impact of playtime on well-being likely depends on the player's underlying motivations and the extent to which the game supports need satisfaction. In this context, the player's experience (PX) serves as a moderating factor: when individuals are intrinsically motivated and derive enjoyment from gameplay, increased playtime is more likely to be positively associated with well-being [20].

Recently, Self-Determination Theory has been increasingly explored in the context of digital game development [43,53,60]. Grounded on SDT, [30] introduced the concept of Radiant Patterns—design patterns intended to support player well-being by satisfying their basic psychological needs. These patterns were conceived as the antithesis of dark patterns. However, as research on radiant patterns progressed [41], it became clear that there was a need to make these patterns more practical and easier to operationalize.

4 Revisiting and Reshaping Game Radiant Patterns

While the existing literature has a solid point on deceptive patterns and how they affect users [7,28,29], it remains scattered when it comes to identifying and presenting positive and ethically designed *actionable* approaches. There are design patterns that work for the construction of specific games, such as for serious games [3], reflective games [44], and empathy games [14].

Particularly addressing DPs in games, [30] first proposed a concept of design patterns that seeks to stimulate the positive potential of human beings, mitigate adverse effects, and called them Radiant Patterns. Their goal was to help fulfill humans' basic psychological needs through design, contributing to the individual's physical and mental health by proposing alternatives that stimulate the player's well-being. [30] proposes the definition:

> *Radiant Game Design Patterns (RPs) are patterns intentionally used by a developer in the creation of digital games to preserve the player's well-being, prevent negative experiences, and support the satisfaction of the player's basic psychological needs.*

This definition was initially intended to serve as an antithesis to the concept of Game Dark Patterns defined by [59], while also referring to intentional design practices in game development that address essential aspects of player well-being and the fulfillment of intrinsic psychological needs. While useful for contrast, we argue that this framing is ultimately too limiting for the full potential of Radiant Patterns. It was a straightforward definition committed to a player-centered, ethically-driven approach to safeguarding psychological well-being, and could provide a positive framework for ethical and positive game design. For

instance, in an expert opinion report for the Kommission für Jugendmedienschutz (KJM; in English, Commission for the Protection of Minors in the Media) [21], RPs were mentioned as a way to support the designation of positive patterns, intended to counteract dark patterns and offer game developers an alternative to guide players into a state of well-being. They were also considered a starting point to avoid unethical practices in game design in the Grand Research Challenges in Games and Entertainment Computing in Brazil - GranDGamesBR 2020–2030 [9].

However, their use remains theoretical, as do many other proposals for countermeasures for DPs. The nature of the RP's definition makes their practical operationalization difficult. Key terms such as "preserve the player's well-being" and "prevent negative experiences" remain broad and somewhat open to individual interpretation, making it challenging for designers to apply or assess them consistently. Besides, the goal of preventing negative experiences—although they can harm players—is an ideal that may not always align with the inherent challenges and emotional range often integral to engaging gameplay. Furthermore, the definition offers limited guidance on empirically verifying that a pattern truly "supports the satisfaction" of basic psychological needs, or how to resolve the inevitable design trade-offs where the satisfaction of one aspect might affect another.

The absence of actionable clarity and quantifiable criteria renders designers unable to convert the noble aspiration of Radiant Patterns into tangible, verifiable design choices, thereby underscoring the need for a more refined and actionable framework. This is supported by studies that applied the concept, Santos [41], showed that the original concept was challenging to implement due to the absence of clear operational parameters.

So, building directly upon the definition established by [30], we propose a **refined definition of Game Radiant Patterns** that keeps the fundamental principles of intentionality and well-being priority, but brings more specificity. It focuses on avoiding a decrease in SDT needs, acknowledges developer goals, minimizes predictable risks, and provides more precise criteria for evaluation (e.g., does it decrease autonomy, relatedness, or competence? Does it utilize known problematic patterns?). It's also more actionable for designers, giving them negative constraints and a preventative focus.

Thus, the refined definition we propose is:

> *Radiant Game Design Patterns (RPs) are interaction patterns intentionally selected and implemented during the digital game design process to allow the pursuit of development goals while ensuring foundational player well-being is not significantly undermined. Specifically, RPs avoid decreasing the player's autonomy, relatedness, and competence. An RP represents a conscious design choice focused on minimizing predictable risks of psychological harm and avoiding design patterns known to consistently frustrate these basic psychological needs, thereby establishing ethical boundaries for player experience.*

Instead of "aiming to satisfy," the new definition shifts the emphasis to a more preventative function that includes preserving current levels of need satisfaction from being undermined by design choices. Hence, the improbable goal of avoiding all negative experiences is replaced with more specific goals: "minimizing predictable risks of psychological harm" and "avoiding design patterns known to frustrate these basic psychological needs consistently." This focuses on anticipated adverse effects directly associated with psychological harm and SDT frustration, giving designers concrete criteria based on known risks and established anti-patterns.

Therefore, the new definition does not replace the ethical core of the original, but rather *puts it into practice*, providing the clarity and practical structure needed to make it more applicable as a concept for ethically guided game design. The concept evolves from a basic moral stance to a more detailed and actionable design guideline. While the original definition of Radiant Patterns was more conceptual, the revised version is more practical and achievable.

To imply that being 'not dark' does not inherently make a pattern 'radiant.' Our refined definition emphasizes a proactive stance: Radiant Patterns are not just about avoiding the frustration of basic psychological needs (which many dark patterns explicitly cause), but about somewhat stimulating well-being, which is not significantly undermined while allowing for the pursuit of development goals. This means an RP is *actively* designed around principles that support (or at least do not detract from) autonomy, competence, and relatedness in a constructive way.

5 Crafting Radiance: A Guide to Propose Radiant Patterns

Following the refinement of the definition of radiant patterns, a format is proposed for documenting and instantiating these patterns, as a means of operationalizing the definition and making them more accessible to the industry. To develop a functional approach to applying Radiant Patterns in game development, we propose a structure to create a Radiant Pattern that follows our refined definition, as game design patterns must have clear structures that distinguish them from one another to provide game designers with additional tools to influence gameplay [22].

To achieve that, we rely on [6] proposal that builds upon [2]'s patterns definition: "*Each pattern describes a problem which occurs over and over again in our environment, and then describes the core of the solution to that problem, in such a way that you can use this solution a million times over, without ever doing it the same way twice*". Björk and Holopainen extend this idea through the lens of game design, defining game design patterns as "*semi-formal, interdependent descriptions of commonly reoccurring parts of the design of a game that concern gameplay.*"

We then identified patterns commonly used in games that promote player well-being, based on the new Radiant Pattern definition introduced in this

work. These patterns were identified through the gameplay experiences of the research team, which consisted of two students (one undergraduate and one post-graduate). Each pattern was then structured and validated by an experienced researcher in the field of HCI with a background in game research.

To develop a radiant pattern model, the research team identified components that would make the pattern as operationable as possible. This would ensure that, when other researchers replicate the model, no questions or doubts would arise about how to apply it. Initially, the pattern followed [22] model of motivational patterns. However, it had to be redesigned to align with the refined definition of a Radiant Pattern grounded on self-determination theory.

The first step in structuring the template was to define the **problem** that a radiant pattern aims to address—typically, situations in digital games that affect players' gameplay. Since deceptive patterns are often used as solutions in such cases, identifying a radiant pattern that achieves the same goal without causing harm can offer a more ethical alternative in a way that could possibly mitigate any negative consequences. However, a pattern is not present in all games; its **context** varies. Therefore, a detailed description of the scenarios and styles in which it can be used is required. After presenting the problem and defining the context, we discussed how the pattern can solve the problem in that context through a **proposed solution**. As the refined definition of a radiant pattern is no longer about satisfying basic psychological needs, but rather about mitigating the frustration associated with these needs, it is necessary to explain how the pattern can promote players' well-being by avoiding a decline in autonomy, relatedness, and competence. Therefore, it is essential to demonstrate how each pattern impacts each basic psychological need individually, providing specific information on how it promotes each need without having a negative effect.

As we design this structure, we consider scenarios in which applying this pattern irregularly could cause the opposite effect of its intended purpose. Therefore, we have identified situations in which the developer must **what for** so that the pattern does not actually harm players. For a more expanded and applicable of how these patterns can either benefit or harm players, we present **examples of implementation**. These examples illustrate situations in which the patterns may be negative—failing to meet the refined definition of RP —or positive, aligning with that refined definition. To finish the template, as the patterns were initially identified through the research team's previous gaming experience, we present the community with a list of **good examples** where this pattern is applied. To better demonstrate and clarify the pattern, this example can be accompanied by an illustration of its practical application.

5.1 Radiant Pattern Structure

Following the validation. We propose a consistent structural template to support the operationalization of this refined R P definition and to guide the identification, development, and documentation of specific Radiant Patterns. This template, detailed in Table 1, outlines the essential components to be considered

and described when articulating an RP. By providing a standardized format, we aim to facilitate clear communication, enable comparative analysis between different patterns, and support the creation of a robust, shareable repository of these beneficial design elements. Each identified Pattern as an RP should follow the same structural template.

Table 1. Structural Template for Radiant Game Design Patterns (RPs)

Template Component	Brief Description
Name	Evocative title for the RP.
Problem	Game design issue the RP addresses.
Context	Recommended scenarios for RP application.
Proposed Solution	Core mechanic/interaction of the RP.
Avoiding Decrease in SDT Needs	How the RP supports Autonomy, Competence, and Relatedness.
Watch For	Potential pitfalls and misapplications.
Implementation Examples (Failing/Matching)	Contrasting correct and incorrect uses.
Good Examples	Illustrative games/features using the RP.

Name. The name of a Radiant Pattern may be a single word or a phrase that describes its fundamental concept. To establish a direct association between the pattern and a positive experience for the player, it is recommended that names be selected that evoke autonomy and a sense of accomplishment. Examples of suitable names include creation, free, and evolution.

Problem. As [2] noted, patterns function as solutions to recurring problem situations. Given that the objective of Radiant Patterns is to promote player well-being, the problems they seek to address pertain to game design issues that have a detrimental effect on players' well-being or that fail to satisfy their basic psychological needs. Consequently, each Radiant Pattern should be accompanied by a description of the game design issue that this pattern aims to address.

Context. Each pattern should be accompanied by a description of recommended scenarios and game styles in which it can be applied. Given that patterns possess distinct characteristics, it is essential to define their recommended use within the game context to ensure the pattern's effectiveness in achieving its intended positive outcomes.

Proposed Solution. As a solution to a recurring problem, each description should present how the RP can effectively address the identified problem. The proposed solution outlines how game mechanics can be implemented to enable the pursuit of development goals while ensuring foundational player well-being.

Avoiding Decrease in Autonomy, Relatedness and Competence. The pattern then continues with a description of how it supports the player's basic psychological needs. The three fundamental needs—autonomy, competence, and relatedness—are addressed individually, with a thorough exposition of how each pattern positively impacts them. These descriptions elucidate the manner in which the needs are supported independently, thereby ensuring that the fulfillment of one need does not interfere with the fulfillment of others.

Watch For. If implemented improperly or outside of the intended context, the efficacy of patterns may be compromised. Therefore, it is essential to include a description of how to minimize predictable risks, avoid frustrating implementations, and highlight examples of poor applications of the pattern. Additional warnings should also be provided to help designers recognize and prevent potential issues.

Example of Implementations: Failing and Matching the RP Criteria. As an extension of the "watch for" section, it is important to present examples of pattern implementations that align with the criteria defined for Radiant Patterns. Likewise, it is necessary to provide examples where the implementation does not meet these criteria. This enables a more in-depth discussion on implementation strategies and key considerations to help mitigate potential obstacles.

Good Examples. As the identified patterns are based on mechanics already present in existing games, it is essential to present a comprehensive list of games that effectively incorporate these patterns. A visual representation of the pattern's practical application within a game can enrich the definition and offer game developers a better understanding of its practical implementation.

5.2 Radiant Pattern Example

In this subsection, we demonstrate the application of the proposed RP structure and present a game mechanic found in existing games that meets the criteria for a Radiant Pattern. This Radiant Pattern is called World Creation. Below, we present its documentation as a Radiant Pattern following the structure (Table 2).

Name. World Creation - Allows players to create and customize scenarios and stages.

Problem. Players may have too little agency, creative paths, or channels of meaningful expression within the boundaries of a game's pre-determined parameters. Inert game worlds can limit long-term play and social interaction based on shared experiences.

Context. Digital games where players interact with game systems and potentially each other. Applicable across various genres (sandbox, platformers, creative suites).

Proposed Solution. Allows players to create, customize, and collaborate on or share scenarios using readily accessible tools and resources, deliberately avoiding

Table 2. Example of a Radiant Pattern: World Creation (Summary)

Template Component	Example - World Creation Summary
Name	World Creation
Problem	Addresses limited player agency, creativity, and expression in pre-defined game worlds.
Context	Digital games (sandbox, platformers, creative suites) allowing player interaction with systems and others.
Proposed Solution	Provides accessible tools for players to create, customize, collaborate, and share scenarios without core functions depending on payment or excessive grind.
Avoiding Decrease in SDT Needs	**Autonomy:** Player control over creations. **Competence:** Mastery of tools, creative achievement. **Relatedness:** Sharing, collaboration, experiencing others' content.
Watch For	Gating core tools behind payment/grind; poorly designed/obfuscated tools; restrictive sharing; insufficient scaffolding for complex tools.
Implementation Examples (Failing/Matching)	**Failing:** Core tools paywalled/grind-locked; buggy tools. **Matching:** Accessible core suite; intuitive tools; easy sharing.
Good Examples	*LittleBigPlanet* (Create Mode), *Super Mario Maker*, *Dreams*.

dependencies on direct payment or excessive repetitive grinding for core creative functions.

Avoiding Decrease in Autonomy, Relatedness, and Competence

Autonomy - World Creation promotes autonomy by giving players plenty of control and freedom (the autonomy to customize their own environments) over their creations and their interaction with the game world. It prevents reducing autonomy by definition.

Competence - The pattern enables players to learn, master tools, be creative, and overcome design challenges (harnessing their creativity, sense of 'accomplishment'). Successfully creating and sharing content builds a sense of competence. It prevents a loss of competence by offering the opportunity for skill development.

Relatedness - Being able to share creations, collaborate, or play others' work (share these creations and invite other players to join, encouraging social interaction) satisfies relatedness needs. Facilitating social connection through collaborative, creative activities helps prevent a decrease in relatedness.

Watch For. A properly designed World Creation pattern would go out of its way not to make fundamental creative tools conditional on excessive grinding or direct payment. These are predicted risks for psychological harm (frustration, exploitation) and are set patterns that necessarily thwart autonomy (loss of liberty to create) and competence (progress feels hindered or artificial). The bottom line for maintaining the World Creation framework as an RP is that fundamental creative tools, basic elements, and the main ability to save and share creations should not be expressly limited by financial constraints or only provided through substituting payment for excessive grind. Monetization should target optional enhancements, platform access, or adjacent community offerings, all while keeping the basic act of creation separate and proficiency-dependent, without creating financial pressure as a barrier to core activity.

Example of Implementations

Implementations Failing to Meet RP Criteria

To gate key components, required mechanisms, or the storage and sharing of works behind boring, repetitive gameplay cycles (grind) or outright financial expenditure. Imposing costs on essential tools would directly hinder autonomy, as players are unable to realize their creative vision and competence freely, given that the capacity to construct is unjustly restricted by external factors unrelated to creative ability.

Poor implementations or Obfuscated Tools. The presentation of creation tools that are poorly designed, buggy, lack basic features (like precise positioning or undo functions), or are poorly documented undermine the player's feeling of competency, making them feel hampered in their ability to realize their ideas because of limitations imposed by the tools instead of limitations imposed by their skill level. Additionally, this reduces autonomy by rendering specific forms of creative expression effectively impossible.

Restrictive Sharing and Discovery Mechanisms. When sharing works is difficult, discovery of the works of others is inadequately supported (e.g., no search/curation), or there are arbitrary restrictions on what or how much can be shared, the relatedness dimension of the pattern is strongly negated. This can be annoying, as many creators' motivations include sharing and communicating with the community. Restricting this diminishes the value of creation.

Insufficient Scaffolding of Sophisticated Tools. Very sophisticated creation environments can overwhelm users if not backed up by solid tutorials, progressive feature unlocking, or overt guidance (scaffolding). Unveiling tremendous complexity without help can consistently lead to frustration and abandonment, precluding competence.

Implementations In line with RP Criteria

Accessible Core Creative Suite. To make a robust set of core creation elements and tools immediately available to the player. Though some extra pieces may be gated behind an upfront barrier, the core capacity for creation

and expression is untethered from unreasonable grinding or monetary expenses. This strongly encourages autonomy and establishes a foundation for competency development.

Intuitive Design and Clear Feedback. Offering user-friendly, responsive tools with clear feedback and adequate tutorials (integrated into the gameplay itself) gives players a sense of empowerment. This directly supports competence by making the tools learnable and usable and minimizing frustration with the interface.

Facilitating Sharing and Community. Emphasize being able to post and discover/play community content easily. Making sharing and discovery easy and satisfying enhances the entire creative ecosystem and avoids the frustrating trend of isolating creators. This might include sharing full games and experiences.

Optional Unlocks By Engaging Play. While fundamental tools ought to be readily available, providing optional cosmetic qualities, upgraded parts, or thematic elements as rewards for participation in many of the game's aspects (which do not solely revolve around grinding) might suit the RP model. This would provide room for progress objectives without restricting required creativity, perhaps augmenting players' senses of competence as they acquire fresh creative possibilities.

Good Examples. The LittleBigPlanet game franchise has a "Create Mode" that allows players to create levels using tools provided by the developers. These levels can be accessed by other players if they are published to the community. Similarly, the Super Mario Maker franchise allows players to create levels using items and resources from other games in the Mario franchise (Fig. 1). Dreams goes even further, allowing players to create entire games and experiences, both visually and musically.



Fig. 1. Super Mario Maker as an example of World Creation Pattern.

6 Challenges, Perspectives and Limitations

As we refined the definition and template of Radiant Patterns, we encountered challenges related to their perception, application, and evaluation. Key issues included variations in player interpretation, measuring their impact on well-being, and balancing industry goals with ethical design. We also highlighted the importance of collaboration between academia and developers for their effective adoption.

6.1 Players Perception on Radiant Patterns

Our perception of radiant pattern is grounded in the intention behind their design: to counteract the harmful effects of deceptive patterns. We expect this mitigation to occur because radiant patterns align with Self-Determination Theory, a fundamental theory for promoting player well-being. However, we acknowledge that each player's perception can affect their experience of these patterns.

First, as [41] highlighted, digital literacy regarding ethics in game development directly affects understanding of these patterns. Player who are familiar with the SDT are more likely to distinguish between positive and deceptive patterns. Second, experienced players may already be familiar with deceptive patterns from their gameplay history. As such, they recognize these patterns as commonly used strategies to engage players, often without fully understanding their consequences.

We cannot be overly confident that these patterns will work in every context which they were designed. Since these patterns were developed empirically, further evaluation is necessary to understand their actual effects, especially when considering different player profiles. Players with certain profiles may react differently to the present of such patterns. Some players may even prefer these patterns, despite the potential negative consequences, while other players didn't even download the game if they know that these patterns are presented in the game.

6.2 A Balance Between Deceptive and Radiant Patterns

We recognize that the terms "deceptive patterns" and "radiant patterns" may reflect a dualistic perspective that associates radiance with virtue and darkness with harm or unethical behavior. However, our goal is not to undermine the industry by proposing these patterns, but rather to create a balance between industry interests and player well-being – an ongoing challenge in developing such design approaches.

Our refined definition comes from the understanding that, while the initial conceptualization of Radiant Patterns positioned them partly as an antithesis to Game Dark Patterns, we believe this perspective warrants refinement. Viewing Radiant Patterns solely as the opposite of Dark Patterns might inadvertently restrict their scope. In our novel proposed perspective, a design could avoid dark patterns yet fail to be 'radiant' if it does not consciously engage with

principles supporting foundational well-being, such as actively considering player autonomy, competence, and relatedness.

Radiant design is more than the absence of harm; it is the presence of thoughtful, ethically-grounded design choices to ensure player well-being is not significantly undermined while pursuing engaging experiences. True radiance implies a constructive, positive orientation that may go beyond simply counteracting manipulative techniques. This nuanced understanding is crucial as we advocate for their broader adoption and exploration. A pattern could be neutral, neither dark nor particularly radiant. Radiant Patterns aim higher, representing a conscious, positive design orientation focused on fostering healthier player-game interactions, rather than merely existing as a negation of harmful practices. This broader, more constructive view allows Radiant Patterns to be a guiding principle for innovative game design, not just a corrective measure.

6.3 The Challenge to Evaluate Players Well-Being

Evaluating the well-being that these patterns may promote in players presents significant challenges. [41] demonstrate that radiant patterns, in contrast to deceptive ones, can raise awareness of ethical principles in digital games. However, evaluating the well-being that these patterns cause in players can be challenging. [41] work shows that radiant patterns, as opposed to deceptive ones, can raise awareness of ethical precepts in digital games. However, it has not yet been possible to investigate the real effects these patterns can have on well-being since they need to be inserted into the context of a game to directly affect gameplay. For gameplay to be complete, other factors affecting the player's experience must be considered, such as visual, sound, and graphic elements. Therefore, there is a need to develop evaluation metrics that focus on radiant patterns in isolation from other gameplay elements.

There is still no methodology to directly evaluate the well-being that these patterns can cause in users. However, it is possible to assess aspects of intrinsic motivation, such as autonomy, competence, and relatedness, which belong to self-determination theory. This can be done mainly through instruments such as the PENS scale [39] and the Intrinsic Motivation Inventory (IMI) [33]. As [8] describes, positive intrinsic motivation and need satisfaction are fundamental to Player Experience because they affect the gameplay moment and the broader impact on player well-being. However, what results from the pattern itself or the experience resulting from all the aspects present in gameplay? We invite academia and the gaming industry to investigate methodologies that can determine whether motivation is genuinely positive due to its connection with the game or the result of using radiant patterns.

6.4 A Radiant Pattern Repository

Radiant Patterns face the challenge of market acceptance, as deceptive patterns are designed strategies that are already well established, widely used, and profitable, despite being ethically inappropriate. They are employed, even when

developers recognize their potential harm [1], which illustrates a complex industry conflict. A next step in the evolution of Radiant Patterns would be to explore how they can generate a return for the game industry, addressing the advantages of Deceptive Patterns and making them more attractive to game developers.

This progress can be collaborative as Radiant patterns can be identified based on players' and researchers' experiences with game mechanics. Applying the template structure proposed in this work makes it possible to systematize the creation and documentation of these patterns. Thus, the Human-Computer Interaction (HCI) community and game developers can collaborate to identify and refine new Radiant Patterns. This collaboration could lead to the development of an open repository of standards, strengthening the creation of a gaming industry guided by ethical principles and committed to promoting player well-being.

6.5 Study Limitations

Although a refined definition and a detailed structure for operationalization is presented, no evaluations with players have yet been conducted to validate the direct effects of Radiant Patterns on well-being. The difficulty in isolating the effects of design patterns from other gameplay elements such as graphics, sound, and narrative also compromises the accuracy of such assessments. Generalizing these results may also be affected, as players' perceptions of the patterns can vary depending on their gaming experience, personal profiles, prior exposure to games, and awareness of the consequences of digital design patterns.

Another important limitation concerns the practical adoption of these patterns by the game industry. Although they offer an ethical alternative to deceptive patterns, Radiant Patterns have not yet demonstrated concrete commercial viability. The adoption of such patterns may not offer game designers an immediate return in terms of their key objectives, including engagement metrics and financial performance.

7 Conclusion

This paper presents a refined definition of radiant patterns and a documentation structure that serve as a template to help game developers as well researches to identify and empirically validate radiant patterns for his operationalization. This structure is exemplified through the implementation of a example of a refined defined Radiant Pattern.

As a refined pattern built upon the Self-Determination Theory, we reinforce that the basic psychological needs of players—autonomy, competence, and relatedness—are not compromised by detrimental design decisions, preserving current levels of need satisfaction as well avoiding adverse effects directly associated with psychological harm. In this work, the authors extend an invitation to the Human-Computer Interaction community to contemplate the development of ethical alternatives to deceptive patterns by employing our systematic tool to collaboratively identify those alternatives.

Acknowledgement. This paper is a partial result of the project Digital Well-Being supported by CNPq (CNPq/MCTI No 10/2023 - UNIVERSAL) under grant number 404559/2023-9.

Disclosure of Interests. The authors declare that they have no competing interests.

References

1. Aagaard, J., Knudsen, M.E.C., Bækgaard, P., Doherty, K.: A game of dark patterns: designing healthy, highly-engaging mobile games. In: CHI Conference on Human Factors in Computing Systems Extended Abstracts, pp. 1–8 (2022)
2. Alexander, C.: A Pattern Language: Towns, Buildings, Construction. Oxford University Press (1977)
3. Arnab, S., et al.: Mapping learning and game mechanics for serious games analysis. Br. J. Edu. Technol. **46**(2), 391–411 (2015)
4. Ballou, N., Deterding, S.: 'i just wanted to get it over and done with': a grounded theory of psychological need frustration in video games. Proc. ACM Hum. Comput. Interact. **7**(CHI PLAY), 217–236 (2023)
5. Baroni, L.A., Pereira, R.: Deceptive patterns under a sociotechnical view. In: Proceedings of the XXIII Brazilian Symposium on Human Factors in Computing Systems, pp. 1–13 (2024)
6. Bjork, S., Holopainen, J.: Patterns in Game Design (Game Development Series). Charles River Media. Inc., Rockland, MA, USA (2004)
7. Bongard-Blanchy, K., Rossi, A., Rivas, S., Doublet, S., Koenig, V., Lenzini, G.: " i am definitely manipulated, even when i am aware of it. it's ridiculous!"-dark patterns from the end-user perspective. In: Proceedings of the 2021 ACM Designing Interactive Systems Conference, pp. 763–776 (2021)
8. Borges, B., Darin, T.: Does it make you shiver under your skin? Stating the importance of psychophysiological measures of well-being in player's experience. In: Workshop sobre Interação e Pesquisa de Usuários no Desenvolvimento de Jogos (WIPlay), pp. 1–11. SBC (2024)
9. Darin, T., Carneiro, N., Miranda, D., Coelho, B.: Challenges in evaluating players' interaction with digital games. In: Forum on Grand Research Challenges in Games and Entertainment, pp. 1–24. Springer (2020)
10. Darin, T., Carneiro, N., Miranda, D., Coelho, B.: Challenges in evaluating players' interaction with digital games. In: Santos, R.P.d., Hounsell, M.d.S. (eds.) Grand Research Challenges in Games and Entertainment Computing in Brazil - GranDGamesBR 2020–2030, pp. 1–24. Springer, Cham (2023)
11. Dhiman, D.B.: Games as tools for social change communication: a critical review. Glob. Media J. **21**, 61 (2023)
12. Fitton, D., Read, J.C.: Creating a framework to support the critical consideration of dark design aspects in free-to-play apps. In: Proceedings of the 18th ACM International Conference on Interaction Design and Children, pp. 407–418 (2019)
13. Fredriksson Friman, E., Zätterlund, O.: The dark patterns of battle passes: Investigating player attitudes to a growing type of microtransaction (2023)
14. Galvão, V.F., Maciel, C., Da Hora Rodrigues, K.R.: How to promote empathy in games? An analysis of the structural elements to be considered in the interaction design. In: Proceedings of the XXII Brazilian Symposium on Human Factors in Computing Systems, pp. 1–12 (2023)

15. Graßl, P., Schraffenberger, H., Zuiderveen Borgesius, F., Buijzen, M.: Dark and bright patterns in cookie consent requests (2021)
16. Gualeni, S., Van de Mosselaer, N.: Ludic unreliability and deceptive game design (2021)
17. Hannula, K.: Deceptive designs in UX: investigating the impact of dark patterns. B.S. thesis (2023)
18. Hsiung, H.H., Lin, C.Y., Zhu, G.Y.: The impact of intellectual capital efficiency on value creation in video game industry—an evidence from Taiwan. J. Infrastruct. Policy Dev. **7**(3) (2023)
19. Jėčius, D., Frestadius, A.: How do players experience a gacha game depending on their perspective as a starting or a veteran player?: A case study of Genshin Impact (2022)
20. Johannes, N., Vuorre, M., Przybylski, A.K.: Video game play is positively correlated with well-being. R. Soc. Open Sci. **8**(2), 202049 (2021)
21. Kammerl, R., Kramer, M., Potzel, K., Wartberg, L.: Förderung von exzessivem Nutzungsverhalten bei Games: Gutachten für die Kommission für Jugendmedienschutz (KJM), December 2023. abrufbar unter einer URL, falls bekannt
22. Lewis, C.: Motivational Design Patterns. Ph.d. dissertation, University of California, Santa Cruz (2013). proQuest ID: Lewis_ucsc_0036E_10365. Merritt ID: ark:/13030/m5bp06nc. https://escholarship.org/uc/item/30j4200s
23. Li, W., Flatla, D.R., Arndt, F.: Divergent deceptions: comparative analysis of deceptive patterns in iOS and Android apps. Behav. Inf. Technol., 1–30 (2025)
24. Luguri, J., Strahilevitz, L.J.: Shining a light on dark patterns. J. Legal Anal. **13**(1), 43–109 (2021)
25. Bhoot, A.M., Shinde, M.A., Mishra, W.P.: Towards the identification of dark patterns: an analysis based on end-user reactions. In: Proceedings of the 11th Indian Conference on Human-Computer Interaction, pp. 24–33 (2020)
26. Malaquias, R., Cardoso, P.: Deception in video games: nine game design patterns. In: International Conference on Design and Digital Communication, pp. 106–120. Springer (2024)
27. Mathur, A., et al.: Dark patterns at scale: findings from a crawl of 11k shopping websites. Proc. ACM Hum. Comput. Interact. **3**(CSCW), 1–32 (2019)
28. Mathur, A., Kshirsagar, M., Mayer, J.: What makes a dark pattern... dark? Design attributes, normative considerations, and measurement methods. In: Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems, pp. 1–18 (2021)
29. Mildner, T., Inkoom, A., Malaka, R., Niess, J.: Hell is paved with good intentions: the intricate relationship between cognitive biases and deceptive design patterns. In: Mitigating Dark Patterns Through Responsible Design, p. 223 (2024)
30. Miranda, D.M., Pontes, R.M., de Gois Ribeiro Darin, T.: It's dark but just a game: towards an ethical and healthy game design practice. In: Proceedings of the 21st Brazilian Symposium on Human Factors in Computing Systems, pp. 1–7 (2022)
31. Niknejad, S., Mildner, T., Zargham, N., Putze, S., Malaka, R.: Level up or game over: exploring how dark patterns shape mobile games. In: Proceedings of the International Conference on Mobile and Ubiquitous Multimedia, MUM 2024, pp. 148–156. Association for Computing Machinery, New York (2024). https://doi.org/10.1145/3701571.3701604
32. van Nimwegen, C., Bergman, K., Akdag, A.: Shedding light on assessing dark patterns: introducing the system darkness scale (SDS). In: 35th International BCS Human-Computer Interaction Conference, pp. 1–10. BCS Learning & Development (2022)

33. Nunes, C., Darin, T.: Cross-cultural adaptation of the intrinsic motivation inventory task evaluation questionnaire into Brazilian Portuguese. In: Proceedings of the XXII Brazilian Symposium on Human Factors in Computing Systems, pp. 1–11 (2023)
34. Nyström, T., Stibe, A.: When persuasive technology gets dark? In: European, Mediterranean, and Middle Eastern Conference on Information Systems, pp. 331–345. Springer (2020)
35. Potel-Saville, M., Da Rocha, M.: From dark patterns to fair patterns? Usable taxonomy to contribute solving the issue with countermeasures. In: Annual Privacy Forum, pp. 145–165. Springer (2023)
36. Prescott, A.T., Sargent, J.D., Hull, J.G.: Metaanalysis of the relationship between violent video game play and physical aggression over time. Proc. Natl. Acad. Sci. **115**(40), 9882–9888 (2018)
37. Przybylski, A.K., Weinstein, N.: Investigating the motivational and psychosocial dynamics of dysregulated gaming: evidence from a preregistered cohort study. Clin. Psychol. Sci. **7**(6), 1257–1265 (2019)
38. Rigby, C.S., Ryan, R.M.: Time well-spent?: Motivation for entertainment media and its eudaimonic aspects through the lens of self-determination theory. In: The Routledge Handbook of Media Use and Well-Being, pp. 34–48. Routledge (2016)
39. Rigby, S., Ryan, R.: The Player Experience of Need Satisfaction (PENS) Model, pp. 1–22. Immersyve Inc. (2007)
40. Ryan, R.M., Deci, E.L.: Self-determination Theory: Basic Psychological Needs in Motivation, Development, and Wellness. Guilford Publications (2017)
41. Santos Filho, L., Sousa Junior, A., Ribeiro, G., Darin, T.: They say that the world was built for two but deceptive patterns say otherwise: a critical game for player empowerment. In: Proceedings of the XXIII Brazilian Symposium on Human Factors in Computing Systems, pp. 1–15 (2024)
42. Sas, M., Denoo, M., Mühlberg, J.T.: Informing children about privacy: a review and assessment of age-appropriate information designs in kids-oriented F2P video games. Proc. ACM Hum. Comput. Interact. **7**(CHI PLAY), 425–463 (2023)
43. Sevinçli, M.C., Aydoğmuş, M.E.: Video games as a part of intervention programs based on self-determination theory. Psikiyatride Guncel Yaklasimlar **14**(2), 207–220 (2022)
44. Shaheen, A., Fotaris, P., Fallahkhair, S.: A systematic review of using reflective design features in game-based learning. In: 15th European Conference on Game Based Learning ECGBL, vol. 2021, pp. 638–645 (2021)
45. Shi, Z., Sun, R., Chen, J., Sun, J., Xue, M.: The invisible game on the internet: a case study of decoding deceptive patterns. In: Companion Proceedings of the ACM Web Conference 2024, pp. 521–524 (2024)
46. Sigman, A.: Screen dependency disorders: a new challenge for child neurology. J. Int. Child. Neurol. Ass. **17**, 119 (2017)
47. Soe, T.H., Nordberg, O.E., Guribye, F., Slavkovik, M.: Circumvention by design-dark patterns in cookie consent for online news outlets. In: Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society, pp. 1–12 (2020)
48. Sousa, C., Oliveira, A.: The dark side of fun: Understanding dark patterns and literacy needs in early childhood mobile gaming. In: European Conference on Games Based Learning, vol. 17, pp. 599–610 (2023)
49. Stevens, M.W., et al.: Global prevalence of gaming disorder: a systematic review and meta-analysis. Aust. New Zealand J. Psychiatry **55**(6), 553–568 (2021)

50. Stockman, C., O'Connell, A., Nottingham, E.: When it's cute but also dark: critical analysis of Farmcute but also Darkille 3's game design in the digital economy. Int. Rev. Law Comput. Technol., 1–23 (2024)
51. Thorsteinsen, K., Vittersø, J.: Now you see it, now you don't: Solid and subtle differences between hedonic and eudaimonic wellbeing. J. Posit. Psychol. **15**(4), 519–530 (2020)
52. Türkay, S., Lin, A., Johnson, D., Formosa, J.: Self-determination theory approach to understanding the impact of videogames on wellbeing during COVID-19 restrictions. Behav. Inf. Technol. **42**(11), 1720–1739 (2023)
53. Tyack, A., Mekler, E.D.: Self-determination theory in HCI games research: current uses and open questions. In: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems, pp. 1–22 (2020)
54. Veiga, E., Silva, N., Gadelha, B., Oliveira, H., Conte, T.: Dark patterns in games: an empirical study of their harmfulness (2025)
55. Waldman, A.E.: Cognitive biases, dark patterns, and the 'privacy paradox'. Curr. Opin. Psychol. **31**, 105–109 (2020)
56. Wijman, T.: Newzoo's year in review: the 2023 global games market in numbers. https://newzoo.com/resources/blog/video-games-in-2023-the-year-in-numbers
57. Willis, L.E.: Deception by design. Harv. JL Tech. **34**, 115 (2020)
58. Yang, Q., Gong, X.: The engagement-addiction dilemma: an empirical evaluation of mobile user interface and mobile game affordance. Internet Res. **31**, 1745–1768 (2021). https://doi.org/10.1108/INTR-11-2020-0622
59. Zagal, J.P., Björk, S., Lewis, C.: Dark patterns in the design of games. In: Foundations of Digital Games 2013 (2013)
60. Zhao, C., et al.: The effects of active video game exercise based on self-determination theory on physical fitness and cognitive function in older adults. J. Clin. Med. **11**(14), 3984 (2022)